

# Project Report

## 1. INTRODUCTION

### 1.1 Project Overview

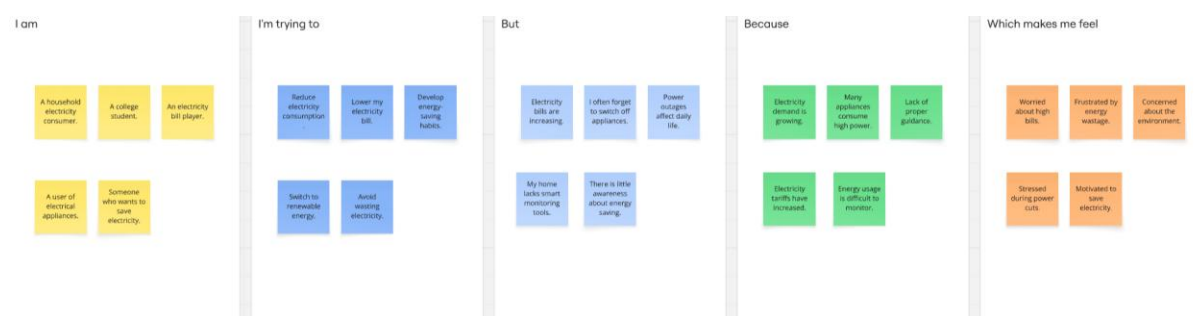
Electricity consumption is a key indicator of economic development, industrial growth, and quality of life. Understanding how electricity is consumed across different regions and time periods helps policymakers, utility providers, and consumers make informed decisions about energy management. This project analyzes electricity consumption patterns across Indian states and regions using interactive visualizations. It examines yearly and monthly consumption trends, regional variations, top and bottom electricity-consuming states, and seasonal changes between 2019 and 2020. The insights generated from this analysis support better energy planning, efficient resource allocation, and the promotion of sustainable electricity usage across the country.

### 1.2 Purpose

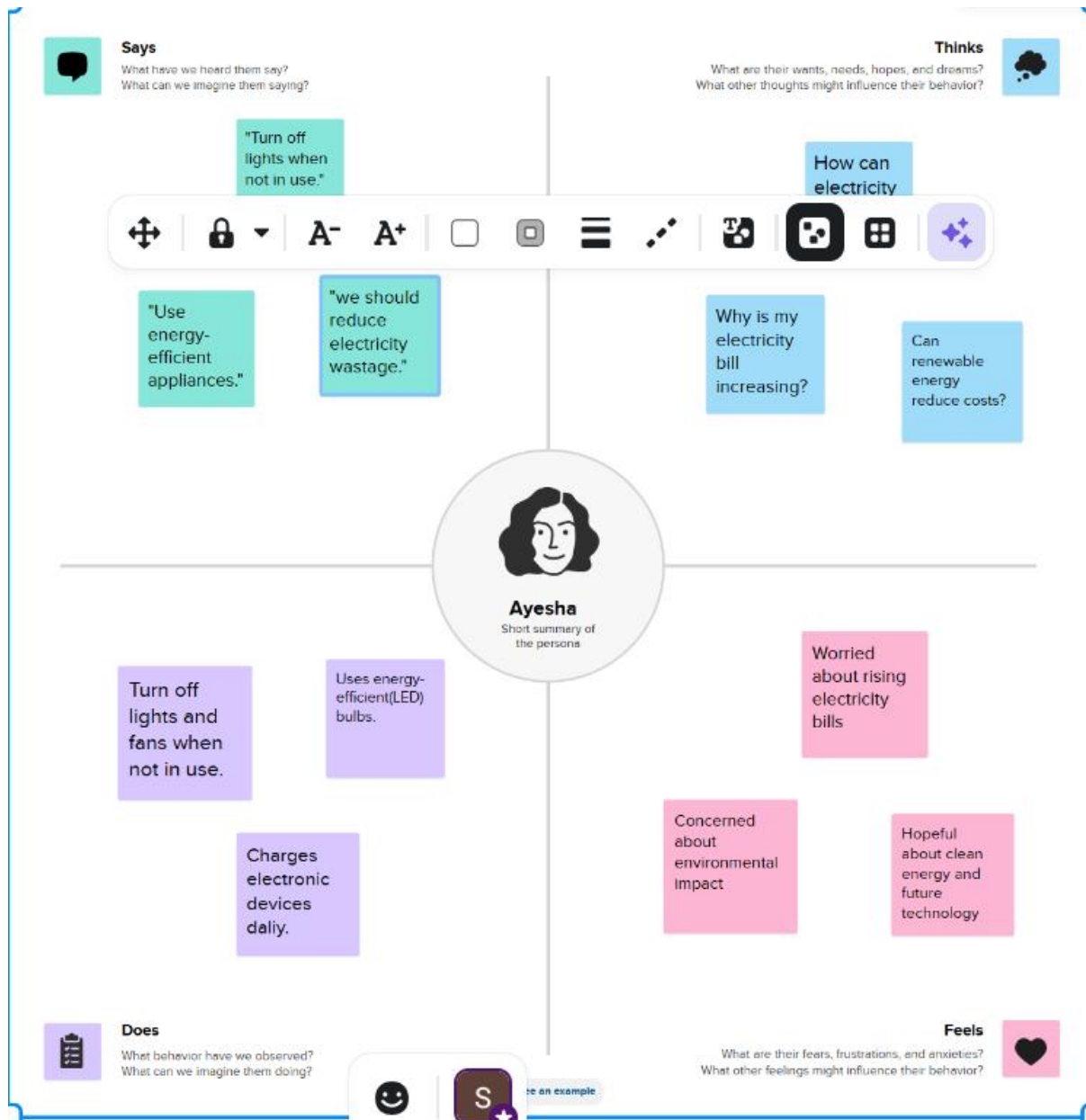
The purpose of this project is to analyze electricity consumption patterns across different states and regions of India. The project aims to identify trends in electricity usage, compare consumption across different time periods, and understand regional variations in energy demand. It also seeks to provide meaningful insights through interactive data visualizations, helping policymakers, utility providers, and researchers make informed decisions for efficient energy management, resource planning, and sustainable electricity consumption.

## 2. IDEATION PHASE

### 2.1 Problem Statement



## 2.2 Empathy Map Canvas



## 2.3 Brainstorming

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Template



## Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

10 minutes to prepare  
 1 hour to collaborate  
 3-8 people recommended

### Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

10 minutes

**Team gathering**

Shah, Aiswhe  
 Shah, Muskan Akshay  
 Shah, Sameer Sushama  
 Chhetrya, Kishan

**Set the goal**

Analyze electricity consumption patterns to identify trends and support efficient energy management.

**Learn how to use the facilitation tools**

Apply facilitation tools to support the analysis of electricity consumption patterns.

[Open article](#)

### Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

5 minutes

PROBLEM

How might we identify the main factors affecting electricity consumption patterns?

Key rules of brainstorming

To run an smooth and productive session

Stay on topic

Deferr judgment

Go for volume

Encourage wild ideas

Listen to others

If possible, be visual

## Step-2: Brainstorm, Idea Listing and Grouping

### Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

**TP**

You can collect a sticky note and move them around the sticky notes to start clustering.

**Person 1**

Use of solar panels  
 Buy energy-efficient appliances  
 Install smart power strips

**Person 2**

Turn off lights when not in use  
 Buy energy-efficient appliances  
 Install smart power strips

**Person 3**

Use of solar panels  
 Buy energy-efficient appliances  
 Install smart power strips

**Person 4**

Use of solar panels  
 Buy energy-efficient appliances  
 Install smart power strips

**Person 5**

Use of solar panels  
 Buy energy-efficient appliances  
 Install smart power strips

**Person 6**

Use of solar panels  
 Buy energy-efficient appliances  
 Install smart power strips

**Person 7**

Use of solar panels  
 Buy energy-efficient appliances  
 Install smart power strips

**Person 8**

Use of solar panels  
 Buy energy-efficient appliances  
 Install smart power strips

### Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

20 minutes

**1. Energy-Efficient Technology**

- Use LED light bulbs
- Buy energy-efficient appliances
- Install smart power strips
- Install smart electricity meters

**2. Renewable Energy**

- Install solar panels
- Use renewable energy sources
- Promote rooftop solar systems
- Support clean energy projects

**3. Energy-Saving Habits**

- Turn off lights when not in use
- Unplug chargers and electronics
- Use natural daylight
- Set air conditioners to energy-saving temperatures

**4. Education and Awareness**

- Educate people about electricity conservation
- Organize school energy-saving campaigns
- Share energy-saving tips
- Encourage community participation

**5. Government and Community Initiatives**

- Offer incentives for energy-efficient homes
- Encourage off-peak electricity use
- Create community energy-saving challenges
- Promote energy-efficient building designs

**TP**

After clustering ideas, keep in mind that you can combine or split clusters to form sub-groups, depending on how many clusters you want to work on.

## Step-3: Idea Prioritization

### 4 Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

20 minutes

**TIP**

Participants can use their devices to jot down ideas; story notes should go on the grid. The facilitator can determine what to copy to the new canvas, leaving the Play on the keyboard.

### After you collaborate

You can export the mural as an image or pdf to share with members of your company who might find it helpful.

**Quick add-ons**

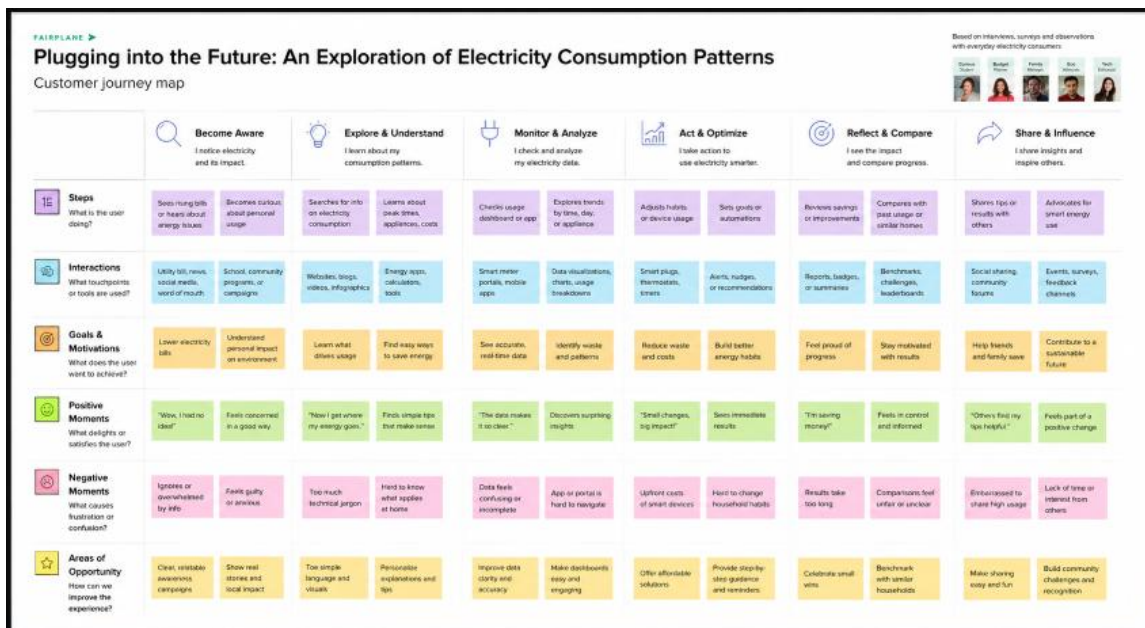
- Share the mural**  
Share a view link to the mural with stakeholders to keep them in the loop about the outcomes of the session.
- Export the mural**  
Export a copy of the mural as a PNG or PDF to share to email, include in slides, or save to your drive.

**Keep moving forward**

- Strategy blueprint**  
Define the components of a new idea or strategy.  
[Open the template](#)
- Customer experience journey map**  
Understand customer needs, motivations, and obstacles for an experience.  
[Open the template](#)
- Strengths, weaknesses, opportunities & threats**  
Identify strengths, weaknesses, opportunities, and threats (SWOT) to develop a plan.  
[Open the template](#)

## 3. REQUIREMENT ANALYSIS

### 3.1 Customer Journey Map



### 3.2 Solution Requirement

#### Functional Requirements:

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic)              | Sub Requirement (Story / Sub-Task)                                                                                                  |
|--------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | Data Collection & Extraction from Database | Collect electricity consumption data, Import data from CSV/Excel/Database, Validate records, Store data in database                 |
| FR-2   | Data Preparation                           | Remove duplicate records, Handle missing values, Clean inconsistent data, Format date/time, Create calculated fields                |
| FR-3   | Data Visualization                         | Create bar charts, Line charts, Pie charts, Monthly and yearly consumption graphs, Appliance-wise usage charts                      |
| FR-4   | Dashboard                                  | Display total electricity usage, Show electricity bill estimate, Filter by date/month/year, Interactive dashboard, Export dashboard |
| FR-5   | Story                                      | Generate insights from data, Highlight high-consumption periods, Compare monthly/yearly usage, Suggest energy-saving actions        |
| FR-6   | Performance Testing                        | Test dashboard loading speed, Verify report generation time, Validate data accuracy, Check system response time                     |
| FR-7   | Web Intergration                           | User login, Connect dashboard to database, Publish dashboard on web, Responsive design for mobile and desktop                       |

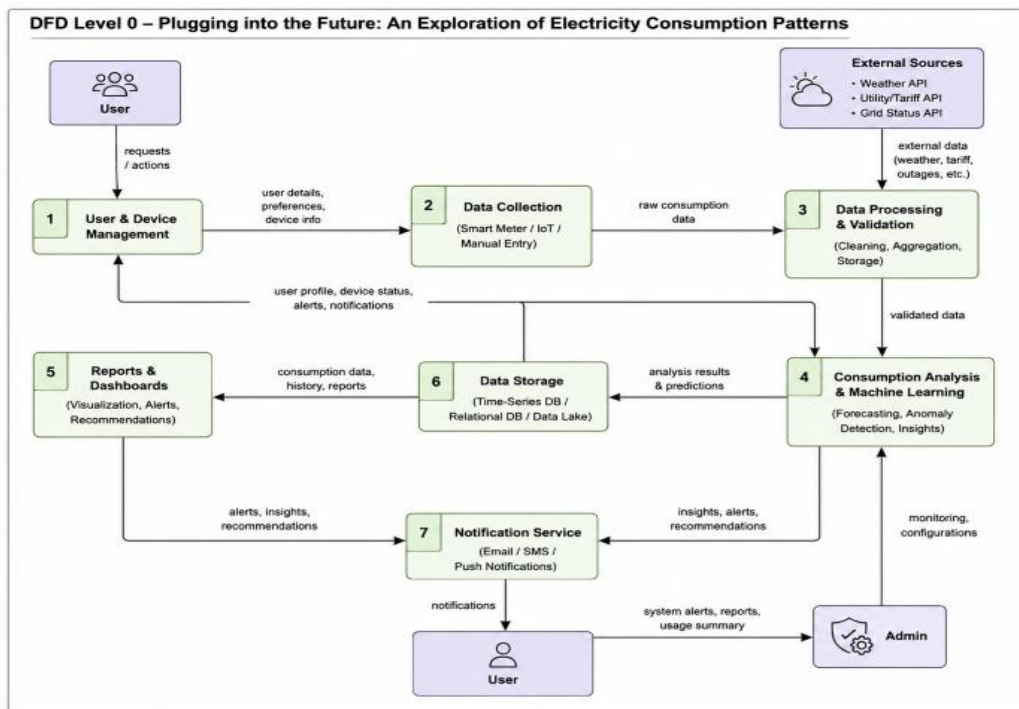
#### Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

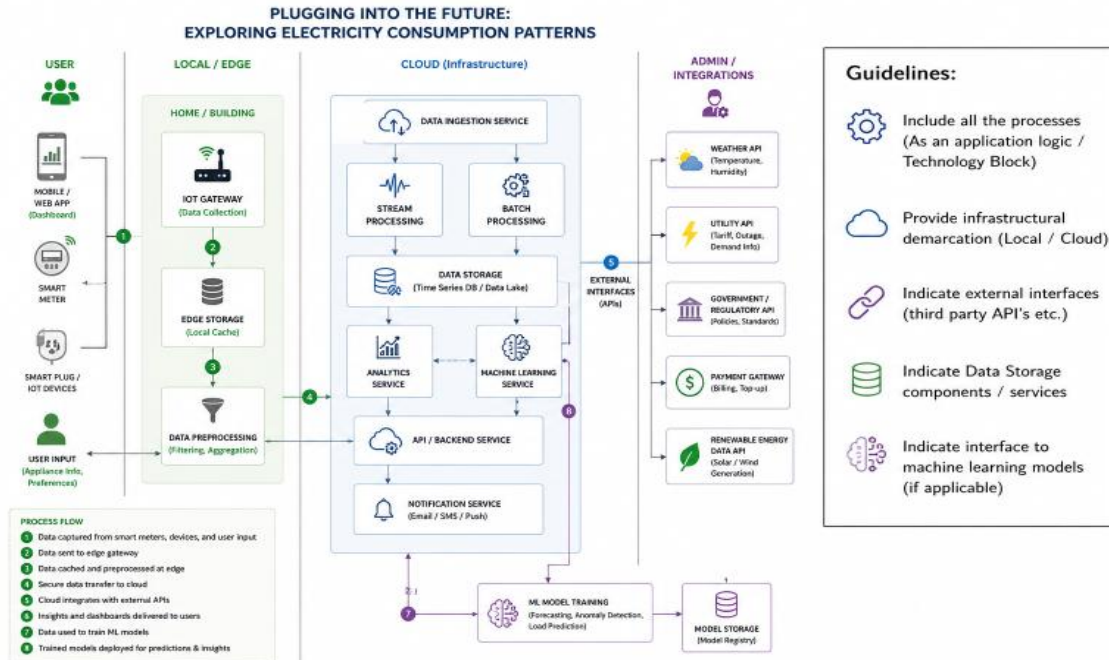
| FR No. | Non-Functional Requirement | Description                                                            |
|--------|----------------------------|------------------------------------------------------------------------|
| NFR-1  | Usability                  | The interface should be simple, easy to understand, and user-friendly. |

|       |                     |                                                                                                                                        |
|-------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| NFR-2 | <b>Security</b>     | User data and electricity usage information should be securely stored and protected.                                                   |
| NFR-3 | <b>Reliability</b>  | The system should provide accurate electricity usage and bill calculations with minimal downtime.                                      |
| NFR-4 | <b>Performance</b>  | The application should load quickly and display data within a few seconds.                                                             |
| NFR-5 | <b>Availability</b> | The system should be available 24/7 with minimal downtime so users can access electricity consumption data anytime.                    |
| NFR-6 | <b>Scalability</b>  | The system should support an increasing number of users and larger volumes of electricity consumption data without performance issues. |

### 3.3 Data Flow Diagram

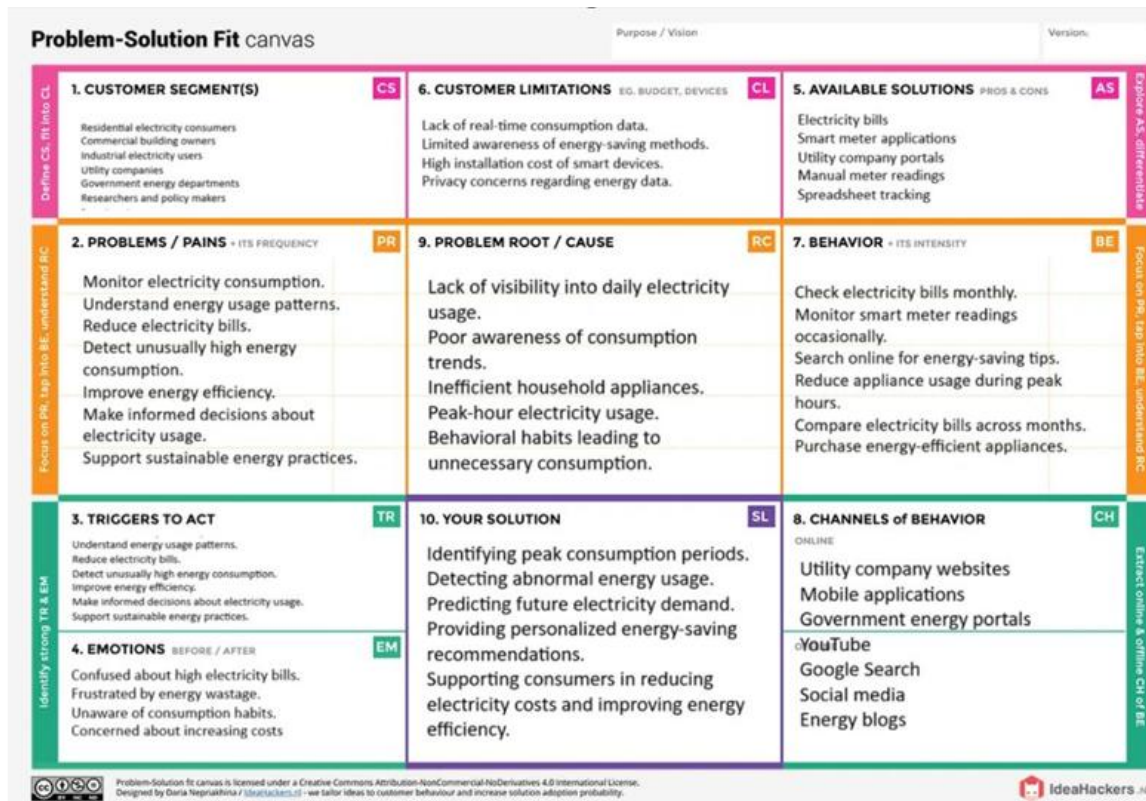


### 3.4 Technology Stack



## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit



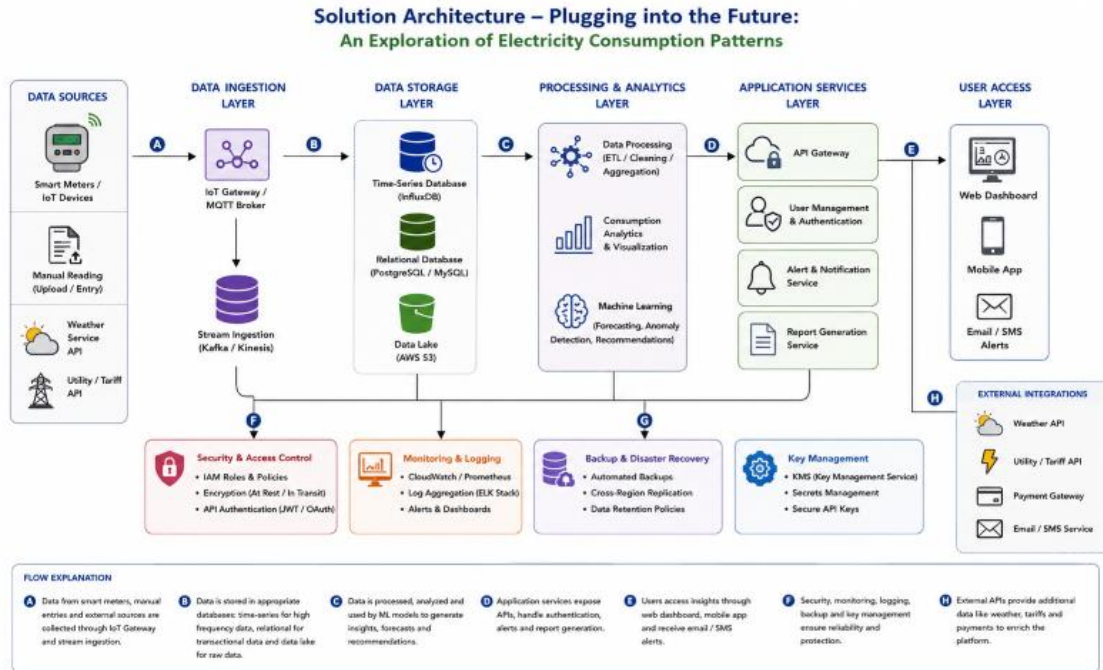


#### 4.2 Proposed Solution

| S.No. | Parameter                                | Description                                                                                                                                                                                                                                                                                |
|-------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | Problem Statement (Problem to be solved) | Electricity consumption patterns vary across states, regions, months, and years. There is a need to analyze this data to identify trends, seasonal variations, regional differences, and consumption changes before and after the lockdown for better energy planning and decision-making. |
| 2.    | Idea / Solution description              | Develop an interactive Tableau dashboard to visualize electricity consumption using charts, maps, filters, and dashboards. The solution provides insights into state-wise, region-wise, monthly, quarterly, and yearly electricity usage to support data-driven decisions.                 |
| 3.    | Novelty / Uniqueness                     | Combines multiple analyses such as state-wise, region-wise, Top N & Bottom N, quarter-wise, metro city usage, and before/after lockdown comparison in a single interactive dashboard with dynamic filters.                                                                                 |
| 4.    | Social Impact / Customer Satisfaction    | Helps government agencies, electricity boards, researches, and consumers understand energy usage patterns, improve energy conservation, optimize resource allocation, and support sustainable electricity management.                                                                      |
| 5.    | Business Model (Revenue Model)           | The dashboard can be offered as a Business Intelligence solution to power distribution companies, government departments, and energy consultants through subscription-based analytics or customized reporting services.                                                                    |
| 6.    | Scalability of the Solution              | The solution can be expanded by integrating live electricity data, smart meter data, renewable energy information, weather data, and predictive analytics for future electricity demand forecasting.                                                                                       |



## 4.3 Solution Architecture



## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                           | Story Points | Priority | Team Members         |
|----------|-------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------|--------------|----------|----------------------|
|          |                               |                   |                                                                                                             |              |          |                      |
| Sprint-1 | User Registration             | USN-1             | As a user, I can create an account using my email and password to access the electricity monitoring system. | 2            | High     | Shaik Ayesha         |
| Sprint-1 | User Registration             | USN-2             | As a user, I can verify my email to activate my account.                                                    | 1            | High     | Syed Salma Sulthana  |
| Sprint-1 | Login                         | USN-3             | As a user, I can securely log in using my registered email and password.                                    | 2            | Low      | Shaik Muskaan Aliyah |

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                          | Story Points | Priority | Team Members         |
|----------|-------------------------------|-------------------|------------------------------------------------------------------------------------------------------------|--------------|----------|----------------------|
| Sprint-1 | Dashboard                     | USN-4             | As a user, I can view my electricity consumption on a dashboard with daily, weekly, and monthly summaries. | 2            | Medium   | Chaitanya Rayachoty  |
| Sprint-2 | Smart Meter Integration       | USN-5             | As a user, I can connect my smart meter or IoT device to automatically collect electricity usage data.     | 1            | High     | Shaik Ayesha         |
| Sprint-2 | Consumption Analytics         | USN-6             | As a user, I can analyze my electricity usage through charts and graphs.                                   | 3            | High     | Syed Salma Sulthana  |
| Sprint-2 | Alerts & Notifications        | USN-7             | As a user, I receive notifications when my electricity usage exceeds a predefined limit.                   | 2            | Medium   | Shaik Muskaan Aliyah |
| Sprint-3 | Energy Saving Tips            | USN-8             | As a user, I receive personalized recommendations to reduce electricity consumption.                       | 3            | High     | Chaitanya Rayachoty  |

#### Project Tracker, Velocity & Burndown Chart: (4 Marks)

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|-------------------------------------------------|------------------------------|
| Sprint-1 | 10                 | 3 Days   | 27 Jun 2026       | 29 Jun 2026               | 20                                              | 29 Jun 2026                  |
| Sprint-2 | 20                 | 2 Days   | 30 Jun 2026       | 01 Jul 2026               | 10                                              | 01 Jul 2026                  |

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|-------------------------------------------------|------------------------------|
| Sprint-3 | 10                 | 3 Days   | 02 Jul 2026       | 02 Jul 2026               | 10                                              | 04 Jul 2026                  |
| Sprint-4 | 20                 | 3 Days   | 07 Jul 2026       | 07 Jul 2026               | 10                                              | 07 Jul 2026                  |

**Velocity:**

**Total Story Points: 40**

**Total Sprint Duration: 11 Days (27 June – 7 July)**

**Average Velocity (Story Points per Day):**

$$\text{Velocity} = \frac{\text{Total Story Points}}{\text{Total Duration}} = \frac{40}{11} = 3.64 \approx 4 \text{ Story Points/Day}$$

## 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

| S.No. | Parameter               | Screenshot / Values                                                                                                                                                                  |
|-------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | Data Rendered           | Electricity Consumption dataset successfully loaded into Tableau/Power BI. Dataset includes State, Region, Year (2019-2020), Month, Quarter, Consumption Units, and Metro City data. |
| 2.    | Data Preprocessing      | Removed null values, corrected data types, formatted Date fields, removed duplicates, standardized State/Region names, and prepared the data for visualization.                      |
| 3.    | Utilization of Filters  | Filters used: Year, State, Region, Month, Quarter, Metro City, Top N/Bottom N.                                                                                                       |
| 4.    | Calculation fields Used | Total Consumption, Average Consumption, Regional Consumption, Quarter-wise Consumption, Year-wise Consumption, Top N & Bottom N calculations.                                        |

## 5. Dashboard design

No of Visualizations / Graphs – 12 (2019 State Consumption, 2020 State Consumption, Total Consumption, Usage By Region, Top N & Bottom N, Month-wise Consumption, Region-wise Consumption, Before & After Lockdown, Region-wise State Usage, Quarter Usage, Metro City Usage, Usage by Year).

## 6 Story Design

No of Visualizations / Graphs – 12, covering all the above analyses in a logical sequence with insights.

## 7. RESULTS

### 7.1 Output Screenshots

2019 State Consumption



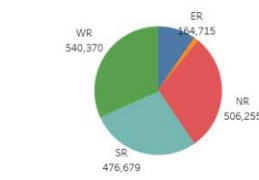
2020 State Consumption



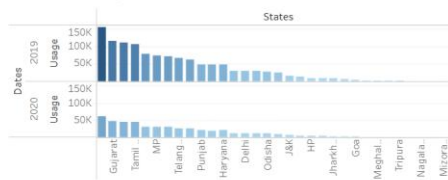
Year Wise Consumption in Region



Total Region Consumption



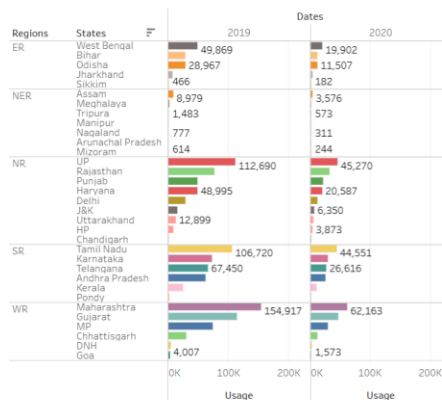
Total Consumption



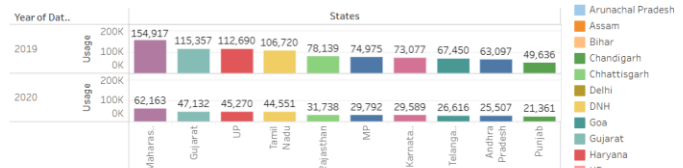
Dashboard 2



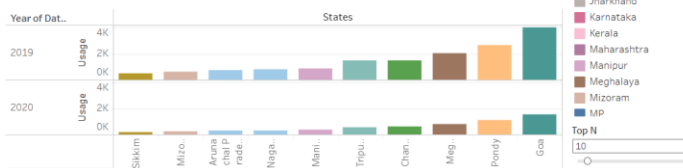
Regionwise State Consumption



Top N



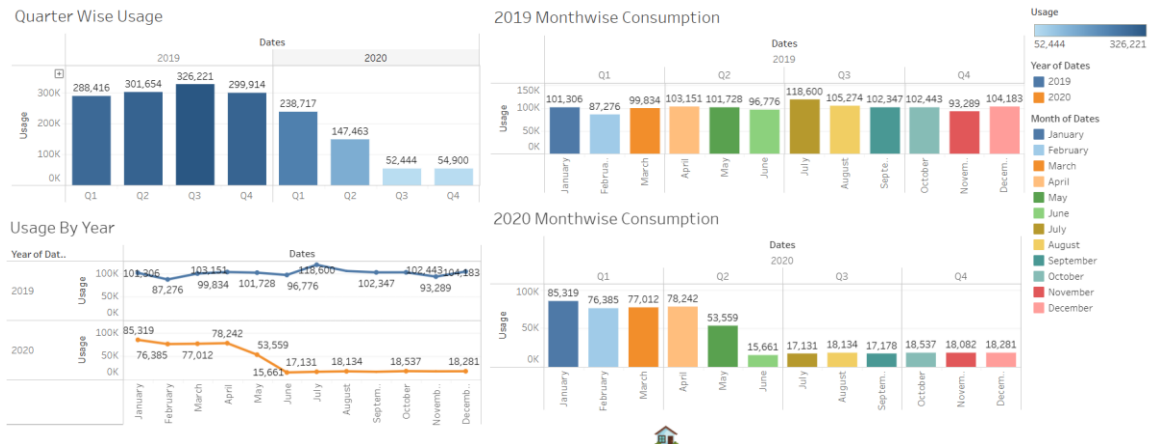
Bottom N



Dashboard 1

Dashboard 3





**Dashboard :**

[https://public.tableau.com/views/dashboard\\_1\\_17831683824880/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/dashboard_1_17831683824880/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

[https://public.tableau.com/views/dashboard\\_2\\_17831684809370/Dashboard2?:language=en-US&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/dashboard_2_17831684809370/Dashboard2?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

[https://public.tableau.com/views/dashboard\\_3\\_17831685994440/Dashboard3?:language=en-US&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/dashboard_3_17831685994440/Dashboard3?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

**Story:**

[https://public.tableau.com/views/storyonelectricityconsumptioninindia/StoryonElectricityConsumptioninIndia?:language=en-US&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/storyonelectricityconsumptioninindia/StoryonElectricityConsumptioninIndia?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

## 1. ADVANTAGES & DISADVANTAGES

### Advantages

- Helps in understanding product placement performance across different store locations and categories.
- Improves decision-making by providing data-driven insights using interactive dashboards.
- Easy identification of high-performing and low-performing products.
- Visualization makes complex data simple and easy to understand.
- Helps businesses optimize shelf space and increase sales efficiency.
- Supports better marketing and inventory planning.

### Disadvantages

- Depends heavily on the quality and accuracy of electricity consumption data; missing or incorrect data can lead to misleading insights.
- Requires knowledge of Tableau and data visualization techniques to effectively interpret dashboards and trends.
- Limited real-time monitoring if the dashboard is not connected to live electricity consumption data sources.
- Insights generated by the dashboard require human analysis and domain expertise for decision-making.
- Large datasets and complex visualizations may reduce dashboard performance and increase loading time.
- Historical data analysis may not accurately predict future electricity demand without advanced forecasting models.
- Variations in data collection methods across states or regions may affect the consistency and reliability of comparisons.
- The dashboard focuses on electricity consumption analysis and does not directly identify the underlying causes of changes in energy usage, such as weather conditions, economic factors, or policy changes.

## 2. CONCLUSION

This project successfully analyzes product placement performance using Tableau visualizations. By cleaning, organizing, and visualizing the dataset, meaningful insights were derived regarding product category performance, sales trends, and placement effectiveness. The interactive dashboard helps users easily explore KPIs and understand how placement impacts sales. Overall, the project demonstrates how data visualization can support better business decisions and improve retail performance.

## 3. FUTURE SCOPE

**Integration of real-time electricity consumption data** to provide live dashboard updates and continuous monitoring.

**Implementation of machine learning models** to forecast future electricity demand and consumption patterns.

**Expansion of the analysis** by incorporating factors such as weather conditions, population growth, industrial activity, and renewable energy usage.

**Development of advanced analytics** to identify consumption clusters, detect anomalies, and optimize energy distribution.

**Creation of a mobile-friendly dashboard** to enable easy access to electricity consumption insights anytime and anywhere.

**Automation of data collection, preprocessing, and reporting** to improve efficiency and reduce manual effort.

**Integration with smart meter and IoT technologies** for more accurate and real-time energy monitoring.

**Support for sustainable energy planning** by analyzing renewable energy consumption and recommending energy-saving strategies.

## 11. APPENDIX

**Source Code (if any)**

**Dataset Link:**

[https://drive.google.com/file/d/1JxIkHNwXxjFztKq7ad0\\_KtkukCqTckNy/view](https://drive.google.com/file/d/1JxIkHNwXxjFztKq7ad0_KtkukCqTckNy/view)

**GitHub & Project Demo Link:**

<https://github.com/Ayeshashaik0425/plugging-into-future-electricity->

<https://drive.google.com/file/d/1K46CMdA5apXu94zSUbf2hH6lE7R8Z4gA/view?usp=sharing>



